

Bhanu Prakash Bathini

Atlanta, GA | (000) 000-0000 | your.email@example.com | [Portfolio](#) | [GitHub](#) | [Hugging Face](#)

EDUCATION

Northwest Missouri State University

Maryville, MO

Master of Science in Applied Computer Science

Aug. 2023 – Dec. 2024

- Coursework: Machine Learning, Database Systems, Data Mining, Cloud Computing, Software Engineering
- Certification: Claude Certified Architect (CCA) — Foundations, Anthropic (2026)

EXPERIENCE

Mondee

Fremont, CA

AI/ML Engineer

Jan. 2023 – Aug. 2026

- Developed machine learning models using PyTorch and scikit-learn that improved routing accuracy by 18% with precision and recall tracking.
- Built grounded document Q&A with Azure AI Search and GPT-4o; traced retrieval errors and reduced hallucination escalations by 40%.
- Queried relational databases with SQL and performed exploratory data analysis to design AI agents connecting models to FastAPI services.
- Presented analysis with visualizations and documentation to product teams; raised groundedness scores by 35% through prompt refinement and error logging.
- Trained classification models in scikit-learn and PyTorch; reported precision, recall, and failure modes each sprint for business reviewers.

Cosmic Reality

Missouri, USA

AI/ML Engineer

Apr. 2020 – Oct. 2022

- Collected and transformed data with SQL and Python pipelines to create structured tables from thousands of malformed PDF pages.
- Trained sentiment and spam classifiers using scikit-learn; improved accuracy by ~12% after addressing class-imbalance issues.
- Developed classification and time-series models in PyTorch and TensorFlow; logged experiments in MLflow for reproducible analysis.
- Deployed scoring applications with Docker on AWS EC2/S3 and documented setup workflows, shortening handoff time by about 40%.

PROJECTS

Agentic Workflow & Eval Harness | *Python, FastAPI, MLflow, n8n, Make.com, Retell AI*

2023 – 2026

- Built multi-step workflow prototypes with Python and FastAPI that call APIs, score outputs, and log metrics in MLflow for comparison.
- Evaluated machine learning outputs using precision, recall, and custom metrics; automated retry logic for failed pipeline steps.
- Developed automation workflows with n8n and Make.com; applied prompt-engineering techniques to improve tool-to-human handoff reliability.

ParseDoc.ai | *Python, PyTorch, Azure, Hugging Face, n8n*

2024 – 2026

- Extracted structured fields from long PDFs using Python and PyTorch detectors; improved bounding-box recovery by ~15% with labeled synthetic data.
- Integrated extraction pipeline into Azure AI Search so document Q&A retrieves grounded source text with auditable trace logs.
- Published labeled datasets on Hugging Face and documented workflows on GitHub for reproducible exploratory data analysis.

RESEARCH, PUBLICATIONS & VOLUNTEER

Peer review: ACM RecSys; ISCA CATA (Springer) — reviews on method, evaluation, and clarity.

Research: Document layout detection and grounded Q&A (ParseDoc.ai); Hugging Face `dataset_parsecdoc.ai`.

Volunteer: Leo Club charitable member — community service, fundraising, and local outreach.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

ML Platforms: scikit-learn, PyTorch, TensorFlow, MLflow

Data & Analytics: Pandas, NumPy, SQL joins, APIs, FastAPI

Cloud & Tools: Azure AI Foundry, Azure AI Search, Docker, AWS EC2/S3

AI Techniques: BERT fine-tuning, prompt engineering, RAG, precision/recall evaluation